

Statistical Analysis of E-Commerce Data: A Learning Journey

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Spring 2026

1 Introduction

This report documents my learning journey through statistical analysis using a real-world e-commerce dataset (530,104 transactions from a UK-based online retailer). I applied five hypothesis tests using Python (`scipy.stats`) at $\alpha = 0.05$. Each section highlights both the statistical technique and what I learned from applying it to messy, real-world data.

2 Data Preparation — Cleaning Matters

The raw dataset had records with negative quantities and zero prices—artifacts that would break any analysis. I learned that real data requires careful cleaning before any statistical work.

Steps taken:

- Removed records with non-positive `Quantity` or `UnitPrice`
- Created `revenue = Quantity × UnitPrice` as the key metric
- The UK dominates: 485,123 records vs 44,981 from all other countries combined

Key insight: The distributions of `Quantity`, `UnitPrice`, and `Revenue` are all heavily right-skewed with extreme outliers. I learned to use 99th percentile clipping for visualization without affecting the actual statistical tests.

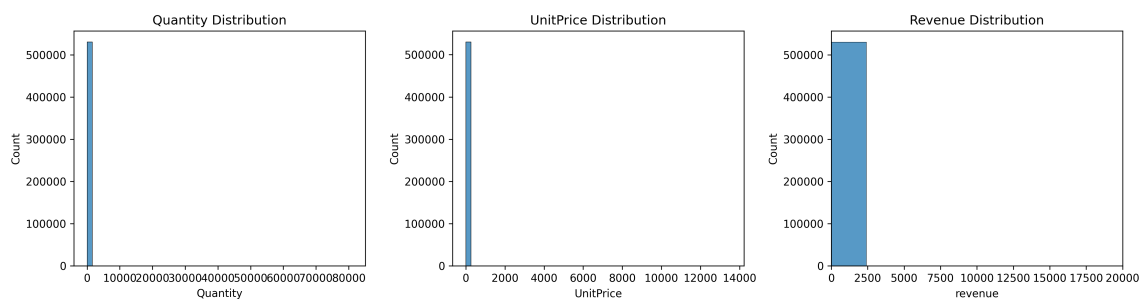


Figure 1: Raw distributions—all three variables are heavily right-skewed.

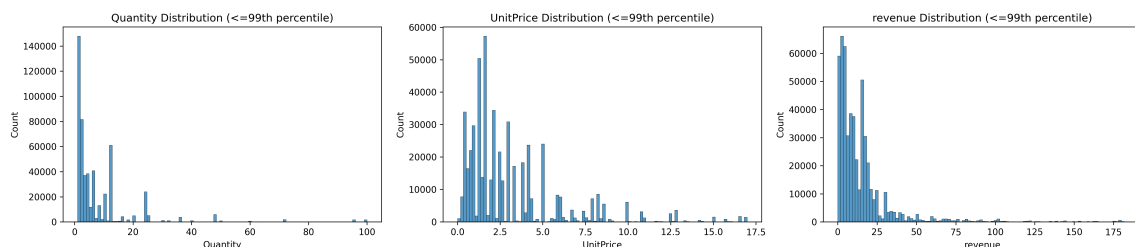


Figure 2: Same distributions clipped at the 99th percentile for readability.

3 Test 1: Normality Testing (Shapiro-Wilk)

What I learned: Before choosing statistical tests, you must check if your data is normally distributed. Many standard tests (t-test, ANOVA, Pearson correlation) assume normality, and violating this can invalidate results.

How it works: Shapiro-Wilk tests H_0 : data comes from a normal distribution. A low p-value means reject normality.

- Sampled 500 revenue values (the full dataset is too large for Shapiro-Wilk)
- Test statistic $W = 0.2061$, $p < 0.001$
- **Result:** Reject H_0 — revenue is NOT normally distributed

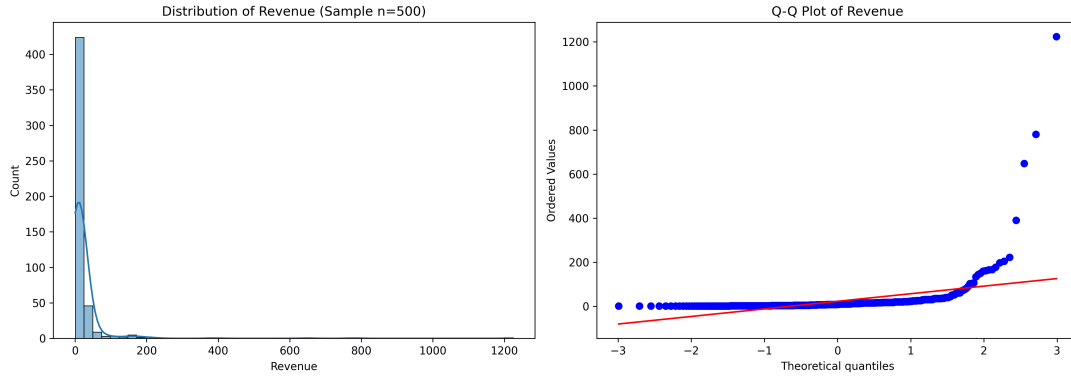


Figure 3: Revenue distribution and Q-Q plot. The data clearly deviates from the diagonal line.

Why this matters: This result guided my choice of non-parametric or robust methods for the remaining tests. I learned that blindly using parametric tests without checking assumptions leads to false conclusions.

4 Test 2: Correlation (Spearman's Rank)

What I learned: Since revenue is not normal, Pearson correlation is inappropriate. Spearman's rank correlation measures monotonic relationships without assuming normality.

Method: Spearman ρ on Quantity vs UnitPrice across all 530K records.

- $\rho = -0.4032, p < 0.001$
- **Result:** Moderate negative monotonic association — as price increases, quantity ordered tends to decrease

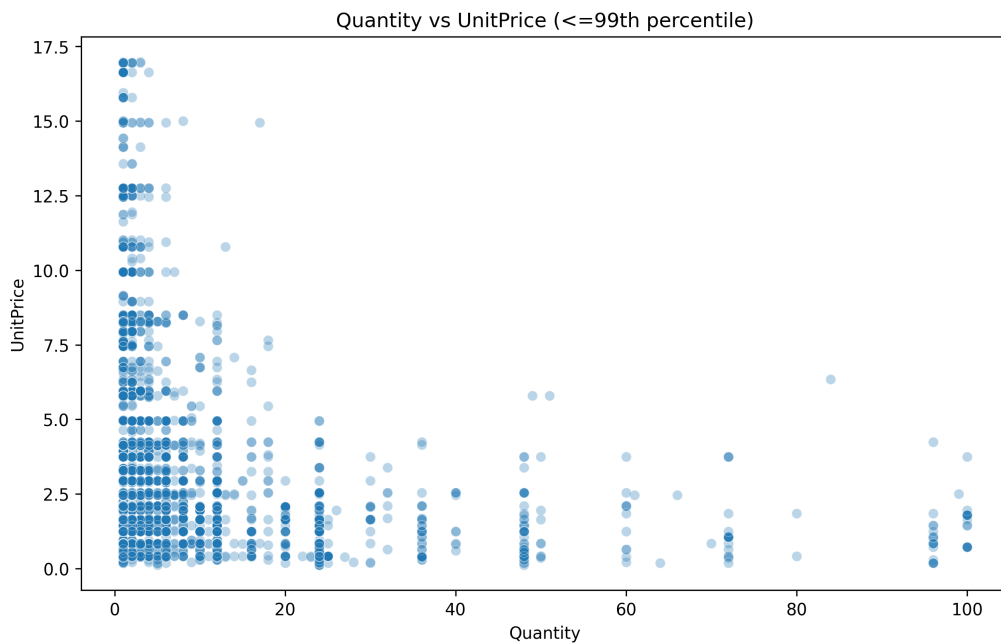


Figure 4: Quantity vs UnitPrice (filtered at 99th percentile, 5,000 sample points).

Why this matters: This confirms basic demand theory with real data. I learned that Spearman is more robust than Pearson when data is skewed, and that "moderate" ($\rho \approx -0.40$) means a noticeable but not perfectly deterministic relationship.

5 Test 3: Two-Sample Comparison (Welch's t-test)

What I learned: Comparing two groups requires checking if their variances are equal first. Levene's test tells you which version of the t-test to use.

Steps:

1. **Levene's test:** $p < 0.001$ — variances are significantly different (UK var = 79,118 vs Other var = 7,811)
2. Because variances are unequal, use **Welch's t-test** (not Student's)

Results:

- UK mean revenue: 18.60
- Other countries mean revenue: 36.49
- $t = -30.8258$, $p < 0.001$
- **Result:** International transactions generate significantly higher revenue per line item

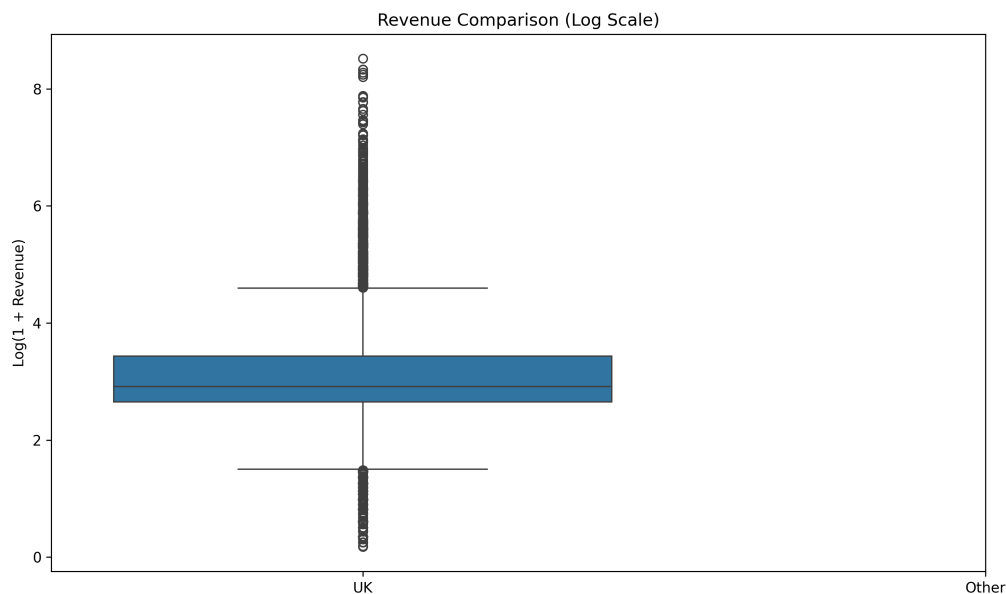


Figure 5: Revenue comparison on log scale. International customers (Other) show higher median revenue.

Why this matters: I learned the importance of the **assumption-checking pipeline**: test normality → test variance homogeneity → choose the correct test. Missing this step would produce unreliable p-values.

6 Test 4: Multiple Group Comparison (One-Way ANOVA)

What I learned: When comparing more than two groups, running multiple t-tests inflates Type I error. ANOVA tests all groups simultaneously.

Approach: Compare mean revenue across the top 5 countries by transaction count: UK, Germany, France, EIRE (Ireland), Spain.

- UK: 18.60, Germany: 25.32, France: 24.95, EIRE: 35.93, Spain: 24.79
- $F = 10.2228$, $p < 0.001$
- **Result:** At least one country's mean revenue is significantly different

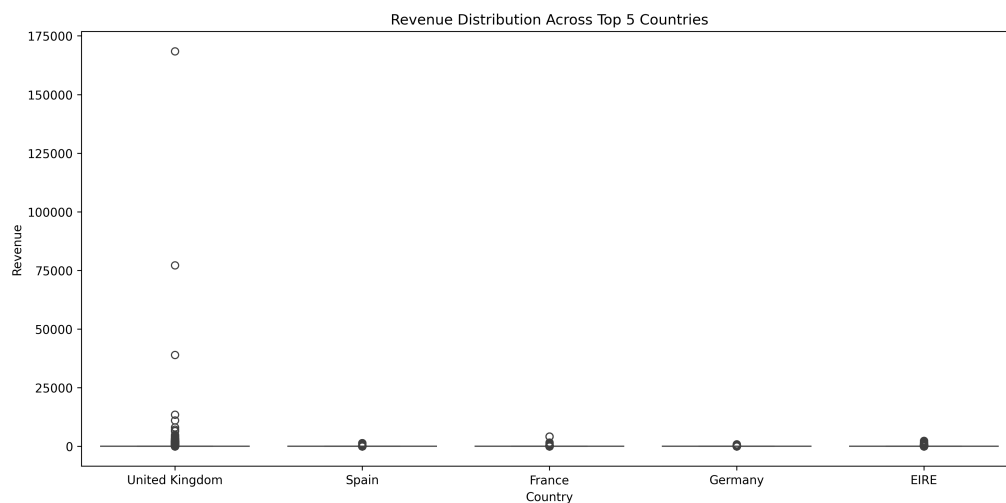


Figure 6: Revenue distribution across top 5 countries. EIRE stands out with higher median revenue.

Why this matters: I learned that ANOVA tells you *if* differences exist but not *where*. Looking at the group means, EIRE (Ireland) has notably higher average revenue (35.93) — this would need post-hoc tests for formal pairwise comparison.

7 Test 5: Categorical Association (Chi-Square Test)

What I learned: Statistical tests aren't just for numerical data. Chi-square tests whether two categorical variables are independent.

Method: Define "high-value purchase" as revenue > 20, then test if country is independent of high-value status.

- $\chi^2 = 3661.10$, $df = 4$, $p < 0.001$
- **Result:** There is a significant association between country and high-value purchase behavior

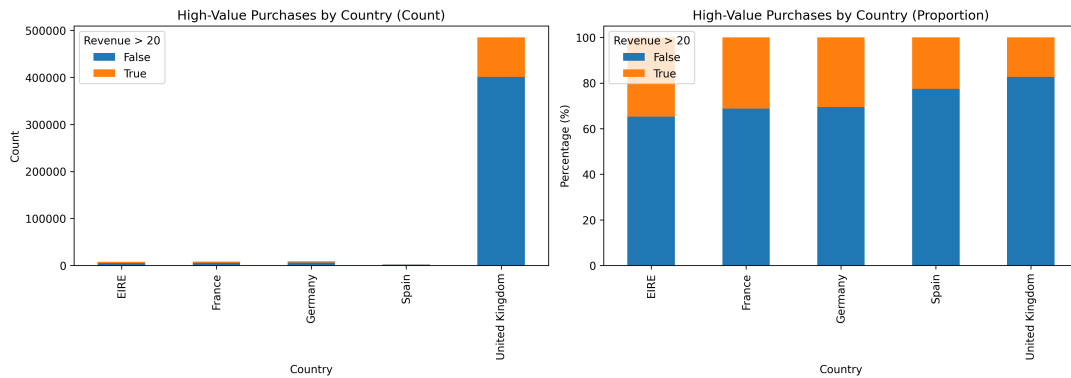


Figure 7: High-value purchases by country: counts (left) and proportions (right). EIRE and France have higher proportions of high-value orders.

Why this matters: I learned that you can convert a continuous variable (revenue) into categories to answer different kinds of questions. The proportion of high-value purchases varies dramatically across countries — a different insight than the mean comparison in Test 3.

8 Conclusion: What I Learned

This project taught me the end-to-end process of statistical analysis on real-world data:

1. **Data preparation** is the most time-consuming but essential step — real data is never clean
2. **Assumption checking** is critical — always test normality and variance homogeneity before choosing a test
3. **Test selection matters:**
 - Shapiro-Wilk for normality
 - Spearman for non-normal correlation
 - Welch's t-test for unequal-variance two-sample comparison
 - ANOVA for multi-group comparison
 - Chi-square for categorical association
4. **Business insights:** International transactions generate nearly double the revenue of UK transactions. EIRE stands out with the highest average revenue and highest proportion of high-value purchases. There is a moderate negative correlation between price and quantity.
5. **Visualization is key:** Outlier clipping, log transforms, and appropriate figure types (histograms, Q-Q plots, boxplots, scatter plots, bar charts) all help communicate findings clearly.

Project repository: <https://github.com/fermow/SBU-Data-Science-2026>